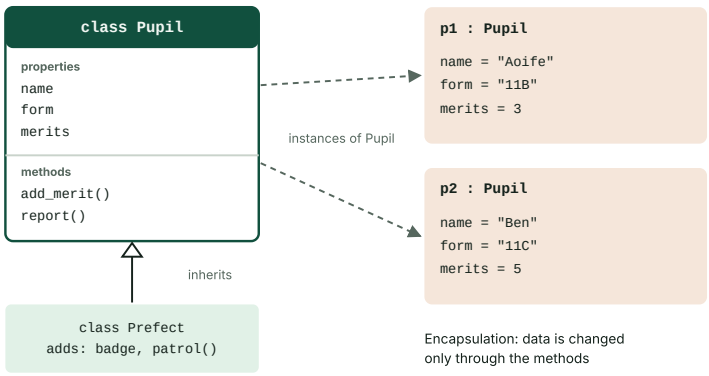
Programming paradigms: procedural and object-oriented

D01 WORKSHEET

Name	Class	Date	Score

I can: Describe procedural programming as imperative, sequential step-by-step instructions · Describe object-oriented programming using objects, classes, properties and methods · Compare the two paradigms in an extended answer



Use this diagram to help you: A Pupil class, two objects created from it, and a Prefect class that inherits from it

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Procedural programming**
- ☐ **Object-oriented programming**
- ☐ **Class**
- ☐ **Object**
- ☐ **Encapsulation**
- ☐ **Inheritance**

A Step-by-step instructions executed in sequence, organised into procedures and functions.

B A blueprint defining the properties and methods of objects.

C A class taking on the properties and methods of another class.

D Keeping an object's data inside it, accessed only through its methods.

E Programs built from self-contained objects that combine data (properties) and methods.

F An instance created from a class with its own data.

B Fill the gaps

Use the word bank · 7 marks

methods | inheritance | procedural | sequence | class | object | properties

In _____ programming the computer follows a _____ of instructions. In object-oriented programming a _____ is a template and an _____ is an instance of it. Data held by an object are its _____ and its actions are its _____. A class that takes on the features of another class uses _____.

C True or false?

Tick one box · 4 marks

	True	False
Procedural languages are described as imperative.	☐	☐
In OOP, an object is a blueprint from which classes are created.	☐	☐
Encapsulation means an object's data can only be changed through its methods.	☐	☐
Object classes cannot be reused in other programs.	☐	☐

D Exam-style questions

17 marks

1 Describe procedural programming. [3]

2 State the difference between a class and an object. [2]

3 In OOP, what is meant by a property and a method? [2]

4 Explain one benefit of inheritance. [2]

5 Describe encapsulation. [2]

6 Describe procedural programming and object-oriented programming. (QWC) [6]

E Quick quiz

Circle one answer · 5 marks

1 Which describes procedural programming? [1]

A Programs built from interacting objects

B Step-by-step instructions executed in sequence

C Programs that only use a GUI

D Code that translates itself

2 A class is... [1]

A an instance of an object

B a blueprint for creating objects

C a type of loop

D a syntax error

3 The actions an object can perform are its... [1]

A properties

B methods

C parameters

D indexes

4 Which OOP feature lets a new class reuse an existing class's code? [1]

A Encapsulation

B Inheritance

C Iteration

D Concatenation

5 Hiding an object's data so it can only be changed through its methods is... [1]

A encapsulation

B decomposition

C abstraction

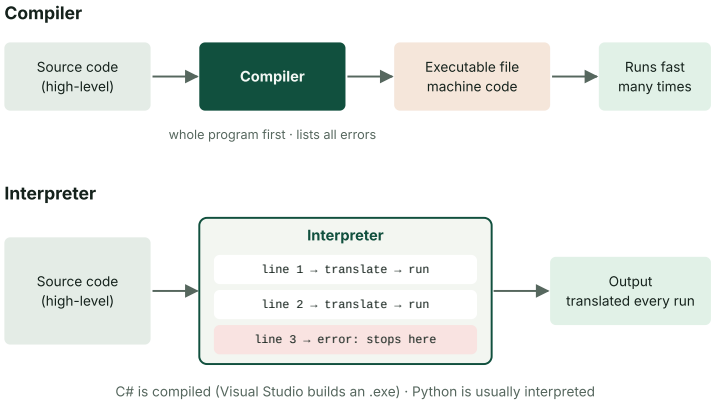
D compilation

Development environments, editing features and translators

D02 WORKSHEET

Name	Class	Date	Score

I can: Describe the features of an IDE and the editing features of a code editor · Explain why high-level code must be translated into machine code · Compare compilers and interpreters



Use this diagram to help you: A compiler translates the whole program into an executable; an interpreter translates and runs one line at a time

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **IDE**
- ☐ **IntelliSense / auto-complete**
- ☐ **Translator**
- ☐ **Machine code**
- ☐ **Compiler**
- ☐ **Interpreter**

A Translates the whole program into machine code before running, producing a separate executable.

B Editor feature that suggests or completes code as it is typed.

C Translates and executes source code one line at a time.

D Integrated development environment: editor, translator, debugger and tools in one application.

E Software that converts source code into machine code: a compiler or interpreter.

F Binary instructions the processor executes directly.

B Fill the gaps

Use the word bank · 6 marks

interpreter | translator | IntelliSense | source | machine | compiler

Programs are written as _____ code in a high-level language. The processor can only run _____ code, so a _____ is needed. A _____ translates the whole program at once and creates an executable file. An _____ translates and runs one line at a time. Editor features such as _____ suggest code as the programmer types.

C True or false?

Tick one box · 4 marks

	True	False
An interpreter produces an executable file that can be run without it.	☐	☐
Compiled programs generally run faster than interpreted ones.	☐	☐
Line numbering helps a programmer find the line given in an error message.	☐	☐
A debugger lets the programmer step through code and watch variables.	☐	☐

D Exam-style questions

12 marks

1 Explain why source code written in a high-level language must be translated. [2]

2 Describe two editing features of an IDE and how each helps a programmer. [4]

3 Describe how a compiler translates a program. [2]

4 Give one advantage of using an interpreter while developing a program. [2]

5 Give one reason why a software company would distribute a compiled program. [2]

E Quick quiz

Circle one answer · 5 marks

1 Which translator converts a whole program into machine code before it runs? [1]

A Interpreter B Compiler
C Debugger D Editor

2 Which IDE feature suggests code and shows parameters while you type? [1]

A Line numbering B IntelliSense/auto-complete
C Code outlining D Breakpoint

3 An interpreter... [1]

A translates one line at a time B creates an .exe file
C only works on binary D removes logic errors

4 The CPU can only execute... [1]

A pseudocode
B machine code
C Python
D flowcharts

5 Collapsing and expanding blocks of code is called... [1]

A code outlining
B syntax highlighting
C compiling
D decomposition

UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · DIGITAL DATA

Binary, units of data and denary–binary conversion

D03 WORKSHEET

Name	Class	Date	Score

I can: Explain why computers store data in binary · Convert denary numbers to 8-bit binary and binary to denary · Use units of data and calculate storage capacities

Place value (each column is double the one to its right)

128	64	32	16	8	4	2	1
0	0	1	0	1	1	1	1

MSB LSB

32 + 8 + 4 + 2 + 1 = 47

Denary to binary: subtract the biggest place value that fits, write 1, move right.
Binary to denary: add the place values that have a 1 under them.

8 bits = 1 byte · 2^8 = 256 values (0 to 255)

Use this diagram to help you: Place values of an 8-bit byte showing 47 as 00101111

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Denary**
- ☐ **Binary**
- ☐ **Hexadecimal**
- ☐ **Carry**
- ☐ **Overflow**
- ☐ **ASCII**

A American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.

B Base-2 number system using 0 and 1.

C Base-10 number system.

D When a result is too large for the available bits, so the extra bit is lost and the result is wrong.

E Base-16 number system (0–9, A–F); a shorthand for binary.

F The digit passed to the next column when a binary column total exceeds 1.

B Fill the gaps

Use the word bank · 6 marks

nibble | bit | byte | megabyte | 255 | 128

A single binary digit is called a _____. Four bits make a _____ and eight bits make a _____. In an 8-bit number the leftmost column is worth _____. The largest number that can be stored in one byte is _____. 1024 kilobytes make one _____.

C True or false?

Tick one box · 4 marks

	True	False
00001111 in binary is 15 in denary.	☐	☐
A byte can store 256 different values.	☐	☐

A nibble is 8 bits.	☐	☐
1 terabyte = 1024 megabytes.	☐	☐

D Exam-style questions

12 marks

1 Explain what is meant by a bit. [2]

2 Convert the denary number 54 into an 8-bit binary number. Show your working. [2]

3 Convert the binary number 10101010 into denary. Show your working. [2]

4 Convert 200 into binary. [2]

5 How many different values can be stored in 4 bits? [1]

6 A disk holds 2 TB. It already stores 1990 GB. Will a 50 GB game fit? Show your working. [3]

E Quick quiz

Circle one answer · 5 marks

1 What is 00010011 in denary? [1]

A 17 B 19
C 21 D 35

2 What is 37 in 8-bit binary? [1]

A 00100101 B 00100011
C 00110101 D 01000101

3 How many bits are in a nibble? [1]

A 2 B 4
C 8 D 16

4 Largest denary number in 8 bits? [1]

A 128 B 255
C 256 D 512

5 How many GB are in 3 TB? [1]

A 3000 B 3072

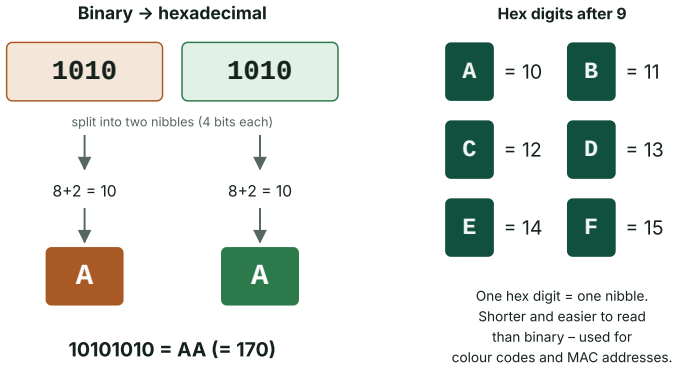
C 1024 D 30 720

Hexadecimal

D04 WORKSHEET

Name	Class	Date	Score

I can: Explain what hexadecimal is and why it is used · Convert between binary and hexadecimal using nibbles · Convert between denary and hexadecimal



Use this diagram to help you: Converting the byte 10101010 to hexadecimal AA by splitting it into nibbles

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Hexadecimal**
- ☐ **Denary**
- ☐ **Binary**
- ☐ **Carry**
- ☐ **Overflow**
- ☐ **ASCII**

A Base-2 number system using 0 and 1.

B When a result is too large for the available bits, so the extra bit is lost and the result is wrong.

C Base-16 number system (0–9, A–F); a shorthand for binary.

D Base-10 number system.

E American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.

F The digit passed to the next column when a binary column total exceeds 1.

B Fill the gaps

Use the word bank · 5 marks

nibble | 16 | A | F | two

Hexadecimal is base _____. The letter _____ represents 15. Each hex digit represents one _____, so a byte is written as _____ hex digits. The binary number 1010 is the hex digit _____.

C True or false?

Tick one box · 4 marks

	True	False
Hex digit B represents 11.	☐	☐
Computers store data in hexadecimal instead of binary.	☐	☐

FF in hexadecimal is 255 in denary.	☐	☐
One hex digit represents 8 bits.	☐	☐

D Exam-style questions

12 marks

1 Convert the binary number 10101010 into hexadecimal. Show your working. [2]

2 Convert the denary number 66 into hexadecimal. Show your working. [3]

3 Convert the hexadecimal number 3B into denary. [2]

4 Convert the hex number C4 into binary. [2]

5 Give one advantage of using hexadecimal instead of binary. [1]

6 Give two uses of hexadecimal in computing. [2]

E Quick quiz

Circle one answer · 5 marks

1 What is 1111 0000 in hex? [1]

A F0 B 0F
C FF D E0

2 Hex digit D represents... [1]

A 12 B 13
C 14 D 15

3 2F in denary is... [1]

A 32 B 45
C 47 D 215

4 Why is hex used? [1]

A Computers can only store hex B It is a shorter, easier-to-read form of binary
C It uses less memory D It encrypts data

5 80 in hexadecimal is... [1]

A 80 B 50

C 48 D A0

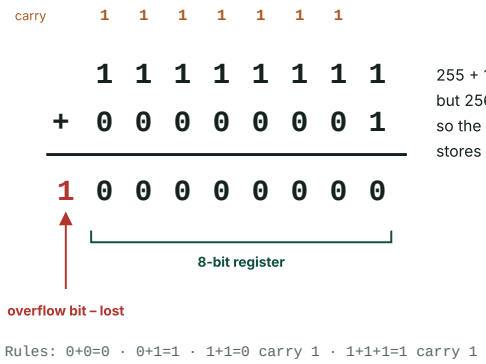
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · DIGITAL DATA

Binary addition and overflow

D05 WORKSHEET

Name	Class	Date	Score

I can: Apply the four rules of binary addition with carries · Add two bytes and check the answer in denary · Explain overflow and its effect on a result



Use this diagram to help you: Adding 11111111 and 00000001 in 8 bits causes overflow

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Carry**
- ☐ **Overflow**
- ☐ **Denary**
- ☐ **Binary**
- ☐ **Hexadecimal**
- ☐ **ASCII**

A The digit passed to the next column when a binary column total exceeds 1.

B When a result is too large for the available bits, so the extra bit is lost and the result is wrong.

C Base-10 number system.

D Base-16 number system (0–9, A–F); a shorthand for binary.

E American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.

F Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

8 | overflow | lost | carry | 255

In binary, 1 + 1 = 0 _____ 1. When the result of an addition is too large for the available bits, _____ occurs. The largest value an _____-bit register can hold is _____. When overflow happens the extra bit is _____, so the stored result is incorrect.

C True or false?

Tick one box · 4 marks

	True	False
1 + 1 + 1 in binary gives 1 carry 1.	☐	☐
Overflow only happens with subtraction.	☐	☐

11111111 + 1 in 8 bits is stored as 00000000.	☐	☐
A carry out of the leftmost column means overflow has occurred.	☐	☐

D Exam-style questions

12 marks

1 Add the binary numbers 00110101 and 00011010. Show the carries. [3]

2 Add 10110110 and 01100101. Show your working. [3]

3 Explain the meaning of overflow. [2]

4 Describe the effect of overflow on the result of an 8-bit addition. [2]

5 Without adding in binary, decide whether 150 + 120 overflows an 8-bit register. Explain. [2]

E Quick quiz

Circle one answer · 5 marks

1 What is 1 + 1 in binary? [1]

A 2 B 11
C 10 D 01

2 00001111 + 00000001 = [1]

A 00010000 B 00011111
C 00001110 D 11110000

3 Overflow occurs when... [1]

A a number is negative B a result is too large for the available bits
C a carry is used D two nibbles are added

4 Which 8-bit addition overflows? [1]

A 100 + 100 B 200 + 50
C 128 + 127 D 200 + 60

5 After overflow the stored result is... [1]

A correct B incorrect because a bit is lost

C always zero

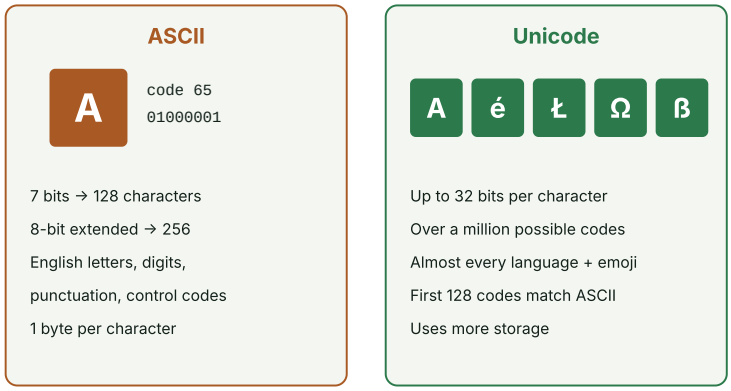
D converted to hex

Character sets: ASCII and Unicode

D06 WORKSHEET

Name	Class	Date	Score

I can: Explain what a character set is · Describe 7-bit ASCII, 8-bit (extended) ASCII and Unicode · Compare ASCII and Unicode



Use this diagram to help you: ASCII compared with Unicode

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **ASCII**
- ☐ **Unicode**
- ☐ **Denary**
- ☐ **Binary**
- ☐ **Hexadecimal**
- ☐ **Carry**

A American Standard Code for Information Interchange: 7-bit (128) or 8-bit (256) character set, mainly English.

B The digit passed to the next column when a binary column total exceeds 1.

C Base-10 number system.

D Character set covering almost all languages and emoji, using up to 32 bits per character; includes ASCII.

E Base-16 number system (0–9, A–F); a shorthand for binary.

F Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

character set | 128 | 8 | more | Unicode

A _____ gives each character a unique binary code. 7-bit ASCII can represent _____ characters. Extended ASCII uses _____ bits. _____ can represent characters from almost every language, including emoji, but uses _____ bits per character.

C True or false?

Tick one box · 4 marks

	True	False
7-bit ASCII can represent 256 characters.	☐	☐
The first 128 Unicode characters match ASCII.	☐	☐

Unicode can represent emoji.	☐	☐
ASCII codes for capital letters are in alphabetical order.	☐	☐

D Exam-style questions

11 marks

1 Expand the acronym ASCII. [1]

2 How many characters can be represented in 7-bit ASCII? Explain. [2]

3 The ASCII code for 'A' is 65. What is the code for 'E'? Give it in denary and binary. [2]

4 Describe two differences between ASCII and Unicode. [4]

5 Explain why a company with customers worldwide would use Unicode. [2]

E Quick quiz

Circle one answer · 5 marks

1 What does ASCII stand for? [1]

A American Standard Code for Information Interchange
B Advanced System Code for Internet Input
C American System for Computer Information
D Automatic Standard Character Index

2 How many characters in 7-bit ASCII? [1]

A 64
B 128
C 256
D 1024

3 Which can represent Chinese characters and emoji? [1]

A 7-bit ASCII
B Extended ASCII
C Unicode
D Hexadecimal

4 A drawback of Unicode compared with ASCII is... [1]

A fewer characters
B more bits per character, so more storage
C it only supports English
D it cannot store digits

5 If 'a' = 97, what is 'c'? [1]

A 98
B 99

C 100 D 67

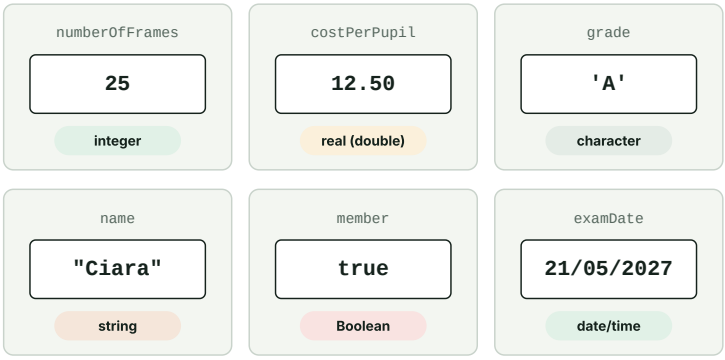
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · DIGITAL DATA

Data types

D07 WORKSHEET

Name	Class	Date	Score

I can: Describe integer, real, character, string, Boolean and date/time data types · Choose and justify a data type for a variable · Use type conversion (casting) in C#



Console.ReadLine() returns a string – convert with Convert.ToInt32() or Convert.ToDouble() first.

Use this diagram to help you: Six variables, each holding a value of a different data type

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Boolean**
- ☐ **Float / real**
- ☐ **Denary**
- ☐ **Binary**
- ☐ **Hexadecimal**
- ☐ **Carry**

A The digit passed to the next column when a binary column total exceeds 1.

B Data type for decimal numbers.

C Base-10 number system.

D Data type with only two values, True or False.

E Base-16 number system (0–9, A–F); a shorthand for binary.

F Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

convert | Boolean | integer | real | string

A whole number is stored as an _____. A number with a decimal part is stored as a _____. A variable that can only be true or false is a _____. Text is stored as a _____. In C#, Console.ReadLine() always returns a string, so Convert.ToInt32() is used to _____ it.

C True or false?

Tick one box · 4 marks

	True	False
A phone number should be stored as an integer.	☐	☐
In C#, Console.ReadLine() returns a string.	☐	☐

A Boolean can store three values.	☐	☐
The price £4.99 should be stored as a real.	☐	☐

D Exam-style questions

10 marks

1 State the most suitable data type for: (a) number of nights booked (b) cost of a booking (c) whether equipment is hired (d) customer name. [4]

2 Describe the Boolean data type. [1]

3 Explain why a race time of 12.84 seconds should be stored as a real. [1]

4 A C# program contains: string score = Console.ReadLine(); int total = score + 5; Explain the error and how to fix it. [2]

5 Describe the difference between a character and a string. [2]

E Quick quiz

Circle one answer · 5 marks

1 Best data type for the number of pupils on a trip? [1]

A Real B Integer
C Boolean D String

2 Best data type for 'Has the pupil paid?' [1]

A Boolean B Character
C Integer D Date/time

3 Which C# method converts "12" into a whole number? [1]

A Convert.ToString() B Convert.ToDouble()
C Convert.ToInt32() D Console.ReadLine()

4 What is the value of 7 / 2 in C# when both are int? [1]

A 3.5 B 3
C 4 D Error

5 Best data type for a customer's surname? [1]

A Character B String

C Real

D Boolean

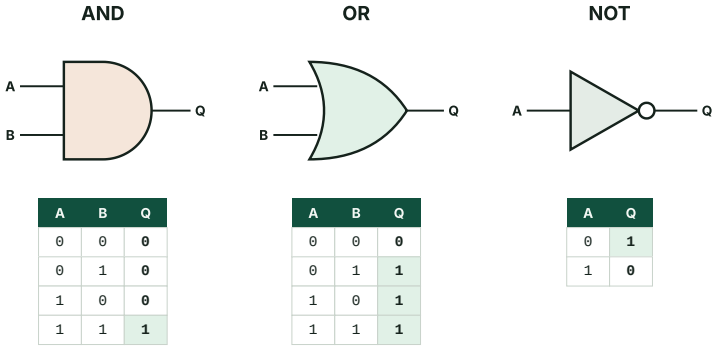
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · DIGITAL DATA

Boolean operators and truth tables

D08 WORKSHEET

Name	Class	Date	Score

I can: Describe the AND, OR and NOT operators · Complete truth tables for single and combined expressions · Use Boolean operators in program conditions



A	B	Q
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Q
0	0	0
0	1	1
1	0	1
1	1	1

A	Q
0	1
1	0

AND: 1 only if both inputs are 1 · OR: 1 if at least one is 1 · NOT: reverses the input

Use this diagram to help you: AND, OR and NOT logic gates with their truth tables

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Truth table**
- ☐ **AND / OR / NOT**
- ☐ **Denary**
- ☐ **Binary**
- ☐ **Hexadecimal**
- ☐ **Carry**

A The digit passed to the next column when a binary column total exceeds 1.

B Table showing the output for every combination of inputs.

C AND is true only if both inputs are true; OR is true if at least one is; NOT reverses the input.

D Base-10 number system.

E Base-16 number system (0–9, A–F); a shorthand for binary.

F Base-2 number system using 0 and 1.

B Fill the gaps

Use the word bank · 5 marks

8 | OR | NOT | AND | truth table

The _____ operator is true only when both inputs are true. The _____ operator is false only when both inputs are false. _____ reverses its input. A _____ shows the output for every combination of inputs. With three inputs there are _____ rows.

C True or false?

Tick one box · 4 marks

	True	False
1 OR 0 = 1	☐	☐
1 AND 0 = 1	☐	☐

NOT 1 = 0	☐	☐
A truth table with two inputs has 4 rows.	☐	☐

D Exam-style questions

12 marks

1 Complete the truth table for A AND B for all four input combinations. [2]

2 Complete the truth table for NOT (P AND Q). [2]

3 How many rows does a truth table with three inputs need? Explain. [2]

4 Complete the output column for (P OR Q) AND R. [4]

5 Write a condition that is True when a mark is between 50 and 100 inclusive. [2]

E Quick quiz

Circle one answer · 5 marks

1 1 AND 1 = [1]

A 0 B 1

C 2 D 10

2 0 OR 0 = [1]

A 0 B 1

C True D Error

3 NOT (1 AND 0) = [1]

A 0 B 1

C 10 D None

4 Which condition is true only for ages 13 to 19? [1]

A age > 13 || age < 19 B age >= 13 && age <= 19

C !(age == 13) D age >= 13 || age <= 19

5 How many rows for 4 inputs? [1]

A 8 B 12

C 16 D 4

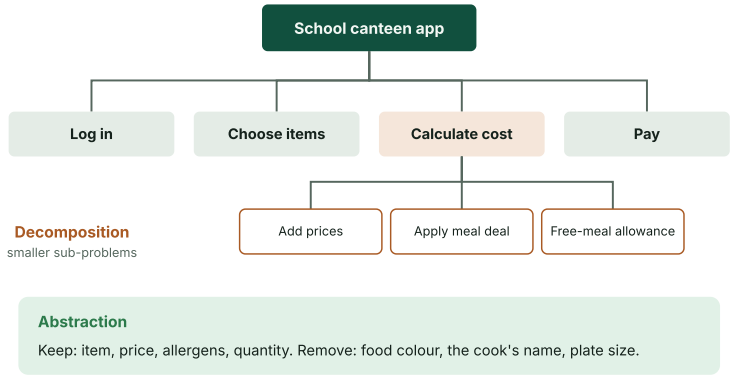
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · DIGITAL DESIGN PRINCIPLES

Computational thinking: abstraction and decomposition

D09 WORKSHEET

Name	Class	Date	Score

I can: Define abstraction and decomposition · Apply both to a real problem · Explain the benefits of each for programmers



Use this diagram to help you: Decomposition of a school canteen app into sub-problems, with an abstraction note

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Top-down design**
- ☐ **Abstraction**
- ☐ **Decomposition**
- ☐ **Algorithm**
- ☐ **Pseudocode**
- ☐ **Flowchart**

A Breaking a complex problem into smaller, more manageable sub-problems.

B Language-independent, English-like instructions describing an algorithm.

C Diagram of an algorithm using standard symbols for start/stop, input/output, process and decision.

D Removing details of a problem that are not needed for the solution.

E Breaking the main task into smaller sub-tasks.

F A step-by-step plan for solving a problem.

B Fill the gaps

Use the word bank · 5 marks

Abstraction | algorithm | Decomposition | sub-problems | structure diagram

_____ means removing details that are not needed. _____ means breaking a problem into smaller _____. A _____ shows top-down design. The final step-by-step solution is an _____.

C True or false?

Tick one box · 4 marks

	True	False
Abstraction means adding extra detail to a problem.	☐	☐

Decomposition lets different programmers work on different parts.	☐	☐
A map is an example of abstraction.	☐	☐
Decomposition makes testing harder.	☐	☐

D Exam-style questions

14 marks

1 Define abstraction. [2]

2 Define decomposition. [2]

3 A cinema wants a booking program. Give two details that could be removed through abstraction. [2]

4 Give two benefits of decomposition when developing a program. [2]

5 Explain abstraction and decomposition and the benefits of each. (QWC) [6]

E Quick quiz

Circle one answer · 5 marks

1 Removing unnecessary detail is... [1]

A decomposition
B abstraction
C iteration
D validation

2 Breaking a problem into sub-problems is... [1]

A decomposition
B abstraction
C compilation
D encryption

3 Which diagram shows decomposition? [1]

A Truth table
B Structure diagram
C Trace table
D Pie chart

4 A benefit of decomposition is... [1]

A programs never have errors
B parts can be developed and tested separately
C no algorithm is needed
D it removes all data

5 The London Underground map is an example of... [1]

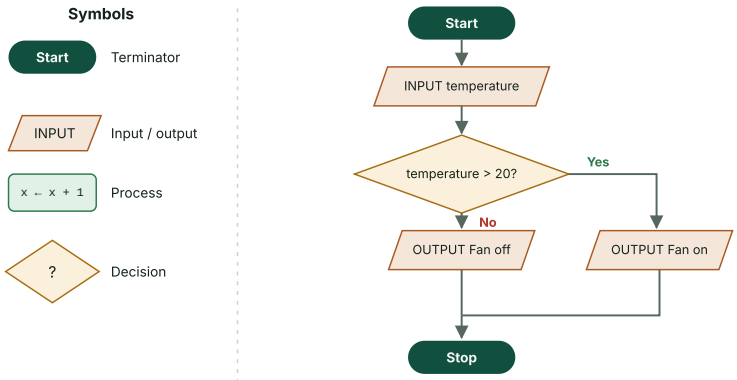
A abstraction
B overflow
C iteration
D a protocol

Designing algorithms: flowcharts and pseudocode

D10 WORKSHEET

Name	Class	Date	Score

I can: Define an algorithm and pseudocode · Use the standard flowchart symbols · Write and refine algorithms in pseudocode using sequence, selection and iteration



Use this diagram to help you: Flowchart symbols and a flowchart that turns a fan on above 20 degrees

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Algorithm**
- ☐ **Pseudocode**
- ☐ **Flowchart**
- ☐ **Top-down design**
- ☐ **Abstraction**
- ☐ **Decomposition**

A Breaking a complex problem into smaller, more manageable sub-problems.

B Language-independent, English-like instructions describing an algorithm.

C A step-by-step plan for solving a problem.

D Diagram of an algorithm using standard symbols for start/stop, input/output, process and decision.

E Removing details of a problem that are not needed for the solution.

F Breaking the main task into smaller sub-tasks.

B Fill the gaps

Use the word bank · 5 marks

algorithm | decision | refinement | parallelogram | Pseudocode

An _____ is a step-by-step solution to a problem. In a flowchart a _____ is shown by a diamond and an input or output by a _____. _____ is English-like and not tied to a language. Improving a design after testing it is called _____.

C True or false?

Tick one box · 4 marks

	True	False
A diamond in a flowchart represents a process.	☐	☐
Pseudocode must follow exact C# syntax.	☐	☐

Algorithms use sequence, selection and iteration.	☐	☐
A terminator shows the start or end of a flowchart.	☐	☐

D Exam-style questions

17 marks

1 Define the term algorithm. [2]

2 Describe what is meant by pseudocode. [2]

3 Name and describe the purpose of two flowchart symbols. [4]

4 Write pseudocode that inputs a temperature and outputs 'Fan on' if it is above 20, otherwise 'Fan off'. [3]

5 Write pseudocode to input five distances in miles, calculate the total and output the total in kilometres (1 mile = 1.609 km). [6]

E Quick quiz

Circle one answer · 5 marks

1 Which flowchart symbol represents a decision? [1]

A Rectangle
B Diamond
C Parallelogram
D Oval

2 Pseudocode is... [1]

A machine code

B a language-independent description of an algorithm

C only used in C#

D a type of flowchart

3 In pseudocode, total ← total + 5 is... [1]

A selection

B an assignment

C a comparison

D an input

4 Which symbol is used for INPUT and OUTPUT? [1]

A Parallelogram

B Diamond

C Terminator

D Arrow

5 Repeating a set of steps is called... [1]

A sequence

B selection

C iteration

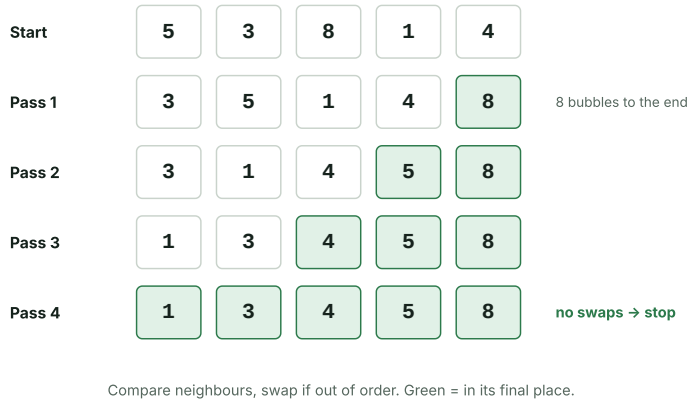
D abstraction

Sorting algorithms: bubble sort and insertion sort

D11 WORKSHEET

Name	Class	Date	Score

I can: Describe how a bubble sort and an insertion sort work · Show each pass of a sort on a list of values · Compare the efficiency of the two sorts



Use this diagram to help you: Bubble sort passes on 5, 3, 8, 1, 4

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Bubble sort**
- ☐ **Insertion sort**
- ☐ **Top-down design**
- ☐ **Abstraction**
- ☐ **Decomposition**
- ☐ **Algorithm**

A Breaking the main task into smaller sub-tasks.

B Builds a sorted list by inserting each item into its correct position.

C A step-by-step plan for solving a problem.

D Breaking a complex problem into smaller, more manageable sub-problems.

E Compares adjacent items and swaps them if out of order, repeating passes until no swaps occur.

F Removing details of a problem that are not needed for the solution.

B Fill the gaps

Use the word bank · 5 marks

pass | bubble | insertion | adjacent | swaps

A _____ sort compares _____ items and swaps them if they are in the wrong order. Each run through the list is called a _____. The sort stops when a pass makes no _____. An _____ sort builds a sorted list by placing each item in its correct position.

C True or false?

Tick one box · 4 marks

	True	False
After the first pass of an ascending bubble sort the largest value is at the end.	☐	☐
Insertion sort needs the data to be sorted before it starts.	☐	☐

A bubble sort compares items that are next to each other.	☐	☐
Bubble sort is very efficient for large lists.	☐	☐

D Exam-style questions

18 marks

1 Describe how a bubble sort orders a list into ascending order. [4]

2 Show the list after each pass of a bubble sort on 7, 2, 9, 4, 1. [4]

3 Describe how an insertion sort works. [3]

4 How does a bubble sort know the list is sorted? [1]

5 Describe the bubble sort and insertion sort and compare their efficiency. (QWC) [6]

E Quick quiz

Circle one answer · 5 marks

1 In a bubble sort, which items are compared? [1]

A First and last
B Adjacent items
C Random pairs
D Middle item and target

2 After pass 1 of an ascending bubble sort on 4 2 5 1: [1]

A 2 4 1 5
B 1 2 4 5
C 2 1 4 5
D 4 2 1 5

3 When does a bubble sort stop? [1]

A After one pass
B When a pass makes no swaps
C When the first item is smallest
D Never

4 Insertion sort works by... [1]

A halving the list
B inserting each item into its place in a sorted sub-list
C swapping the first and last items
D searching for the middle item

5 Which is usually faster on nearly sorted data? [1]

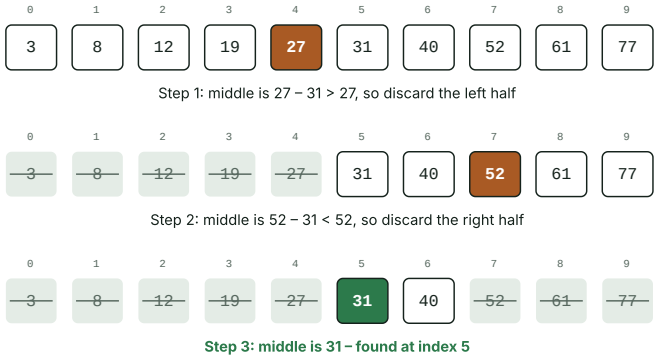
A Bubble sort
B Insertion sort
C Both are identical
D Linear search

Searching algorithms: linear and binary search

D12 WORKSHEET

Name	Class	Date	Score

I can: Describe linear and binary search · Trace a binary search to find or not find an item · Compare the efficiency and requirements of each



Use this diagram to help you: Binary search for 31 in a sorted list, halving the search each step

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Linear search**
- ☐ **Binary search**
- ☐ **Top-down design**
- ☐ **Abstraction**
- ☐ **Decomposition**
- ☐ **Algorithm**

A Breaking a complex problem into smaller, more manageable sub-problems.

B Repeatedly checks the middle of a sorted list and discards half; needs sorted data.

C A step-by-step plan for solving a problem.

D Breaking the main task into smaller sub-tasks.

E Checks each item in turn until the target is found or the list ends.

F Removing details of a problem that are not needed for the solution.

B Fill the gaps

Use the word bank · 5 marks

sorted | binary | linear | faster | middle

A _____ search checks each item in turn. A _____ search only works if the data is _____. It looks at the _____ item and discards half the list each time, so it is much _____ for large lists.

C True or false?

Tick one box · 4 marks

	True	False
Linear search works on unsorted data.	☐	☐
Binary search checks every item one by one.	☐	☐

Binary search needs sorted data.	☐	☐
For a very short list, a linear search is perfectly suitable.	☐	☐

D Exam-style questions

11 marks

1 Describe how a linear search finds an item. [2]

2 Describe how a binary search finds an item. [3]

3 State the condition that must be met before a binary search can be used. [1]

4 Give two advantages of a binary search over a linear search. [2]

5 Trace a binary search for 40 in: 3 8 12 19 27 31 40 52 61 77. Show each middle value. [3]

E Quick quiz

Circle one answer · 5 marks

1 Which search requires sorted data? [1]

A Linear B Binary
C Both D Neither

2 The first item a binary search checks is... [1]

A the first B the last
C the middle D a random item

3 Searching 1 000 items, which is likely fastest? [1]

A Linear B Binary
C Bubble sort D Insertion sort

4 Linear search on [4, 9, 2, 7] for 7 needs how many comparisons? [1]

A 1 B 2
C 3 D 4

5 After checking the middle item and finding the target is larger, a binary search... [1]

A stops B discards the lower half
C discards the upper half D restarts

UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · DIGITAL DESIGN PRINCIPLES

Dry runs, trace tables, data requirements and interfaces

D13 WORKSHEET

Name	Class	Date	Score

I can: Carry out a dry run using a trace table · Predict the output of an algorithm, including loops · Identify data requirements and design a simple user interface

```
a ← 1
WHILE a < 20
    a ← a * 2
END WHILE
OUTPUT a
```

Check the condition every time, including the last one: 32 < 20 is False, so the loop stops and 32 is output – not 16.

a	a < 20 ?	Output
1	True	
2	True	
4	True	
8	True	
16	True	
32	**False**	**32**

Use this diagram to help you: A dry run of a WHILE loop recorded in a trace table

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Dry run**
- ☐ **Trace table**
- ☐ **Desk checking**
- ☐ **Top-down design**
- ☐ **Abstraction**
- ☐ **Decomposition**

A Working through code by hand, line by line, recording variable values.

B Breaking a complex problem into smaller, more manageable sub-problems.

C Reviewing code by hand to find errors before running it.

D Removing details of a problem that are not needed for the solution.

E Table recording the values of variables at each step of a dry run.

F Breaking the main task into smaller sub-tasks.

B Fill the gaps

Use the word bank · 5 marks

dry run | prompts | trace table | logic | data requirements

Working through an algorithm by hand is a _____. The values of each variable are recorded in a _____. Dry runs help find _____ errors. Before coding, the designer identifies the _____, including each variable's data type. A good interface gives clear _____.

C True or false?

Tick one box · 4 marks

	True	False
A dry run needs the program to be compiled first.	☐	☐

Trace tables record variable values as the algorithm runs.	☐	☐
In a nested loop the inner loop runs fully for each pass of the outer loop.	☐	☐
Data requirements only list the program's outputs.	☐	☐

D Exam-style questions

10 marks

1 Explain how a dry run helps when testing a program. [2]

2 What is a trace table? [1]

3 Predict the output of: for (int i = 1; i < 4; i++) { Console.WriteLine(i * i); } [2]

4 Complete a trace table for: total ← 0; FOR n ← 1 TO 4: total ← total + n; NEXT n; OUTPUT total. State the final output. [3]

5 A hire program asks for number of nights, whether equipment is needed and the cost per night. Identify a suitable data type and validation rule for number of nights. [2]

E Quick quiz

Circle one answer · 5 marks

1 What is the output? x ← 3; WHILE x < 30: x ← x * 3; OUTPUT x [1]

A 9 B 27
C 81 D 30

2 A trace table records... [1]

A CPU speed B variable values at each step
C test plan results only D file sizes

3 for (int i = 2; i < 5; i++) Console.Write(i + " "); outputs... [1]

A 2 3 4 5 B 2 3 4
C 1 2 3 4 D 3 4 5

4 A dry run is mainly used to find... [1]

A syntax errors B logic errors
C hardware faults D viruses

5 Which belongs in data requirements? [1]

A Variable names and data types

B The colour of the PC

C The CPU clock speed

D The programmer's name

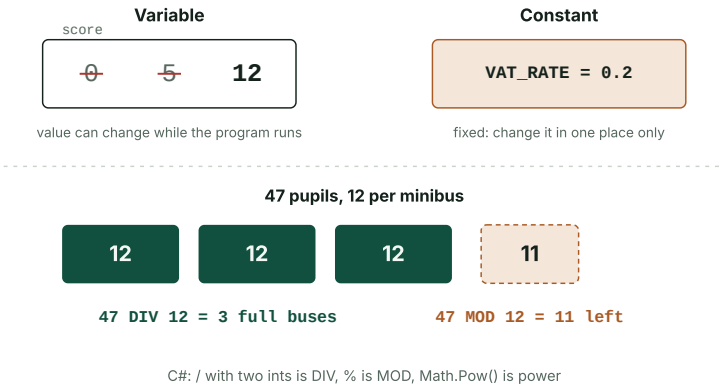
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

Variables, constants, operators, input, output and assignment

D14 WORKSHEET

Name	Class	Date	Score

I can: Explain the difference between a variable and a constant and the benefits of constants · Use input, output and assignment statements · Use arithmetic operators including DIV, MOD and powers



Use this diagram to help you: A variable whose value changes compared with a constant, and DIV and MOD

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Variable**
- ☐ **Constant**
- ☐ **Sequence**
- ☐ **Selection**
- ☐ **Iteration**
- ☐ **Infinite loop**

A A named memory location whose value can change while the program runs.

B Repeating a block of code (loops).

C Statements executed one after another in order.

D A loop whose condition never becomes false, so it never ends.

E A named value that stays the same while the program runs.

F A condition decides which statements run (IF statements).

B Fill the gaps

Use the word bank · 5 marks

constant | assignment | DIV | variable | MOD

A _____ is a named memory location whose value can change. A _____ keeps the same value while the program runs. Storing a value in a variable is called _____. The _____ operator gives the remainder of a division and _____ gives the whole-number part.

C True or false?

Tick one box · 4 marks

	True	False
17 MOD 5 = 2	☐	☐

A constant can be changed by the user while the program runs.	☐	☐
In C#, == is used to compare two values.	☐	☐
Variables should be given meaningful names.	☐	☐

D Exam-style questions

10 marks

1 Describe the difference between a variable and a constant. [2]

2 Give two advantages of using a constant in a program. [2]

3 What is the result of 23 MOD 5 and 23 DIV 5? [2]

4 Write a pseudocode statement that assigns the result of price multiplied by quantity to a variable called total. [1]

5 Write pseudocode to input the length and width of a room and output its area. [3]

E Quick quiz

Circle one answer · 5 marks

1 Which value should be a constant? [1]

A A pupil's score
B The VAT rate
C A running total
D The user's name

2 What is 14 MOD 4? [1]

A 3
B 2
C 3.5
D 1

3 What is 14 DIV 4? [1]

A 3
B 2
C 3.5
D 4

4 Which is an assignment? [1]

A OUTPUT total
B total ← total + 1
C IF total > 10
D INPUT total

5 An advantage of a constant is... [1]

A it changes automatically
B it only needs updating in one place

C it uses no memory

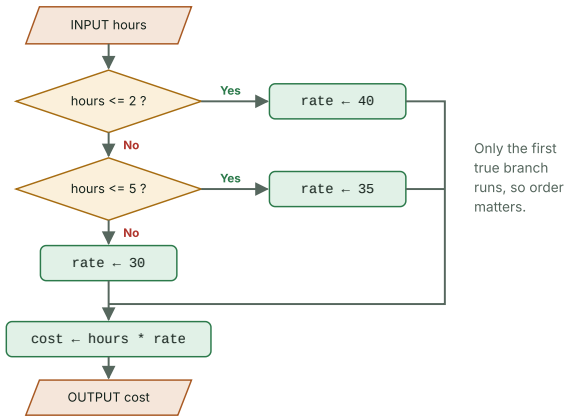
D it speeds up the CPU

Selection: IF, ELSE IF and nested IF

D15 WORKSHEET

Name	Class	Date	Score

I can: Describe sequence and selection · Write IF, IF–ELSE and multi-branch selection in pseudocode and C# · Write tiered-pricing algorithms using single-condition IFs



Use this diagram to help you: Flowchart of an IF, ELSE IF, ELSE price calculation

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Sequence**
- ☐ **Selection**
- ☐ **Variable**
- ☐ **Constant**
- ☐ **Iteration**
- ☐ **Infinite loop**

A Statements executed one after another in order.

B A loop whose condition never becomes false, so it never ends.

C Repeating a block of code (loops).

D A named memory location whose value can change while the program runs.

E A condition decides which statements run (IF statements).

F A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

Selection | condition | ELSE | sequence | nested

_____ allows a program to choose which statements to run. The block runs only when the _____ is True. An _____ branch runs when the condition is False. Placing one IF inside another is a _____ IF. Statements running one after another is _____.

C True or false?

Tick one box · 4 marks

	True	False
In an else if chain, every true branch is executed.	☐	☐
else must have a condition.	☐	☐

Selection uses conditions that evaluate to True or False.	☐	☐
Boundary values should be used to test selection.	☐	☐

D Exam-style questions

12 marks

1 Describe the difference between sequence and selection. [2]

2 What is output when x = 7? IF x > 10 THEN OUTPUT 'A' ELSE IF x > 5 THEN OUTPUT 'B' ELSE OUTPUT 'C' END IF [1]

3 Write pseudocode that inputs a number of tickets and outputs the total: fewer than 5 tickets cost £8 each, 5 or more cost £6 each. [4]

4 Explain why the order of conditions matters in an ELSE IF chain. [2]

5 Give three test values for a program where 1–2 hours costs £40/hour, 3–5 hours £35/hour, over 5 hours £30/hour. Explain your choices. [3]

E Quick quiz

Circle one answer · 5 marks

1 x = 15. IF x > 10 AND x < 20 THEN OUTPUT 'Yes' ELSE 'No' [1]

A Yes B No
C Error D Nothing

2 In C#, which keywords add another condition after an if? [1]

A else B else if
C then D elif

3 int score = 50; if (score > 50) "Merit" else if (score >= 40) "Pass" else "Fail" [1]

A Merit B Pass
C Fail D Merit Pass

4 An IF inside another IF is... [1]

A a loop B a nested IF

C a function

D an array

5 Which construct runs statements in order? [1]

A Sequence

B Selection

C Iteration

D Recursion

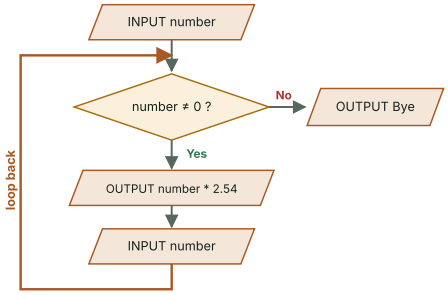
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

Iteration: FOR, WHILE and REPEAT loops

D16 WORKSHEET

Name	Class	Date	Score

I can: Describe bounded (count-controlled) and unbounded (condition-controlled) iteration · Write FOR and WHILE loops including running totals and counters · Use sentinel values and avoid infinite loops



WHILE: unbounded – condition checked before each repeat; 0 is the sentinel

```
for (int i = 0; i < 5; i++)
```

0 1 2 3 4

FOR: bounded – repeats a set number of times. i < 5 gives 0 to 4, not 5.

do ... while (C#) and REPEAT...UNTIL check at the end: always run once.

Use this diagram to help you: Flowchart of a WHILE loop with a sentinel value, and a FOR loop counting

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Iteration**
- ☐ **Infinite loop**
- ☐ **Variable**
- ☐ **Constant**
- ☐ **Sequence**
- ☐ **Selection**

A A condition decides which statements run (IF statements).

B Statements executed one after another in order.

C A loop whose condition never becomes false, so it never ends.

D Repeating a block of code (loops).

E A named memory location whose value can change while the program runs.

F A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

WHILE | iteration | infinite | FOR | sentinel

Repeating code is called _____. A _____ loop repeats a set number of times. A _____ loop repeats while a condition is True. A special input value that ends a loop is a _____. A loop whose condition never becomes False is an _____ loop.

C True or false?

Tick one box · 4 marks

	True	False
A WHILE loop always runs at least once.	☐	☐
Running totals should be initialised before the loop.	☐	☐

for (int i = 1; i < 5; i++) runs with i = 5.	☐	☐
REPEAT...UNTIL checks its condition at the end.	☐	☐

D Exam-style questions

13 marks

1 Describe the difference between a FOR loop and a WHILE loop. [2]

2 What is output? total ← 0; FOR i ← 1 TO 4: total ← total + i * 2; NEXT i; OUTPUT total [2]

3 Explain what an infinite loop is and give one cause. [2]

4 A loop should end when 0 is entered. It uses WHILE inches > 0. Explain the problem and correct it. [2]

5 Write pseudocode that inputs numbers until -1 is entered, then outputs how many numbers were entered and their total. [5]

E Quick quiz

Circle one answer · 5 marks

1 Which loop is count-controlled? [1]

A WHILE B REPEAT UNTIL
C FOR D IF

2 How many times does for (int i = 3; i < 7; i++) run? [1]

A 3 B 4
C 5 D 7

3 x ← 10; WHILE x > 0: x ← x - 3; OUTPUT x [1]

A 0 B 1
C -2 D -1

4 Where should the output of a final total go? [1]

A Inside the loop B After the loop

C Before the loop

D In the condition

5 A loop that never ends is... [1]

A a nested loop

B an infinite loop

C a bounded loop

D a sentinel

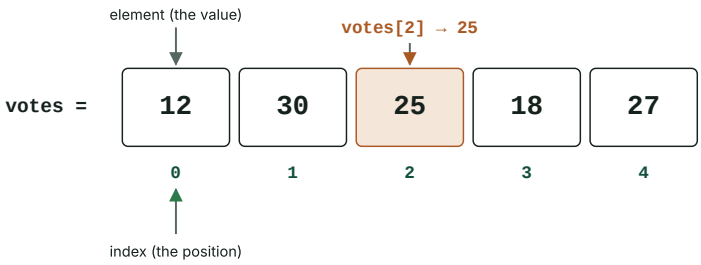
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

One-dimensional arrays

D17 WORKSHEET

Name	Class	Date	Score

I can: Describe an array and its index · Initialise, access, update and loop through an array · Use arrays with running totals, counts, maximum and minimum



Indexes start at 0 · the last index is votes.Length − 1 · loop while i < votes.Length

Use this diagram to help you: An array of five votes showing elements and zero-based indexes

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Array**
- ☐ **Index**
- ☐ **Variable**
- ☐ **Constant**
- ☐ **Sequence**
- ☐ **Selection**

A A data structure holding values of one data type under one name, accessed by index.

B A condition decides which statements run (IF statements).

C The position number of an element in an array (starting at 0).

D Statements executed one after another in order.

E A named memory location whose value can change while the program runs.

F A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

element · index · loop · array · zero

An _____ stores many values of the same data type under one name. Each value is an _____ and its position is its _____. Arrays are _____-indexed, so the first element is at position 0. A _____ is used to process every element.

C True or false?

Tick one box · 4 marks

	True	False
The contents of an array cannot be changed.	☐	☐

The index of an element stores its value.	☐	☐
In a 6-element array, the last index is 5.	☐	☐
All elements in an array normally have the same data type.	☐	☐

D Exam-style questions

17 marks

1 Describe what is meant by an array. [2]

2 An array votes holds [14, 22, 9, 31]. What is the value of votes[2]? [1]

3 Write a statement that changes the fifth element of the array heights to 162. [2]

4 Write pseudocode using a loop to set all 30 elements of the array scores to 0. [3]

5 Write pseudocode to output the total and average of the 7 values in the array cars. [5]

6 Complete an algorithm to find the lowest value in the array votes (8 elements). [4]

E Quick quiz

Circle one answer · 5 marks

1 string[] names = { "Ali", "Bea", "Cal" }; names[1] is... [1]

A Ali
B Bea
C Cal
D Error

2 The index of the first element is... [1]

A 1
B 0
C -1
D Its value

3 Which loop processes every element of a 10-element array? [1]

A for (int i = 0; i < 10; i++)

B for (int i = 1; i < 10; i++)

C for (int i = 0; i <= 10; i++)

D while (i > 10)

4 To find the largest value you should start with... [1]

A largest = 0 always

B largest = the first element

C largest = the last index

D largest = array.Length

5 int[] arr = { 3, 6, 9 }; arr[0] + arr[2] = [1]

A 9

B 12

C 15

D 18

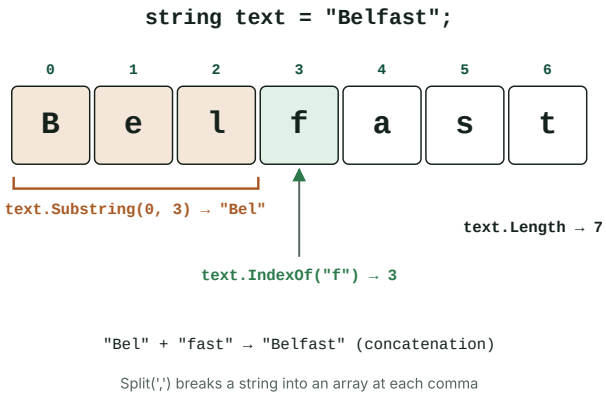
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

String manipulation

D18 WORKSHEET

Name	Class	Date	Score

I can: Use concatenation, Length, indexing and Substring · Split strings and search for characters and substrings · Apply string functions to validation problems



Use this diagram to help you: String indexes, Substring, IndexOf and Length for the word Belfast

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Concatenation**
- ☐ **String function**
- ☐ **Variable**
- ☐ **Constant**
- ☐ **Sequence**
- ☐ **Selection**

A Statements executed one after another in order.

B A named memory location whose value can change while the program runs.

C Joining strings together.

D A function that manipulates text, e.g. length, split, find, substring.

E A condition decides which statements run (IF statements).

F A named value that stays the same while the program runs.

B Fill the gaps

Use the word bank · 5 marks

concatenation | Length | substring | 0 | Split

Joining two strings is called _____. The _____ property gives the number of characters.

Part of a string is a _____. The _____ method breaks a string into parts at a separator.

The first character of a string is at index _____.

C True or false?

Tick one box · 4 marks

	True	False
'Hello'[1] returns 'H'.	☐	☐
IndexOf() returns -1 if the substring is not found.	☐	☐

In C#, a number joined to a string with + is converted to text automatically.	☐	☐
Split(',') removes all the commas and returns one string.	☐	☐

D Exam-style questions

10 marks

1 What is meant by concatenation? [1]

2 Name a function that returns the number of characters in a string and give an example. [2]

3 word ← 'COMPUTER'. What is the substring from index 3 to index 5 inclusive? [1]

4 Explain how Split(',') could be used to process the line "Lisburn,Cat,4". [2]

5 Describe two string functions that could check a password is at least 8 characters and contains @. [4]

E Quick quiz

Circle one answer · 5 marks

1 "Bel" + "fast" = [1]

A Bel fast
B Belfast
C fastBel
D Error

2 "Derry".Length = [1]

A 4
B 5
C 6
D 0

3 "Antrim".Substring(2, 2) = [1]

A nt
B tr
C tri
D ntr

4 "a,b,c".Split(',') gives... [1]

A "abc"
B an array {"a", "b", "c"}
C 3
D {"a,b,c"}

5 "jo@ni.uk".Contains("@") returns... [1]

A true
B false
C 2
D -1

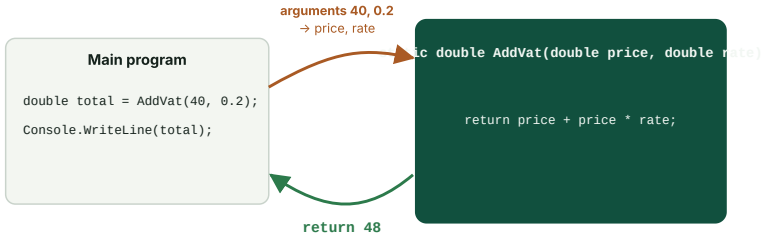
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

Subroutines: user-defined functions and procedures

D19 WORKSHEET

Name	Class	Date	Score

I can: Define subroutine, procedure and function · Write functions with parameters and return values · Explain the benefits of user-defined functions



Write it once · call it many times · test it on its own

A function returns a value; a procedure does not.

Use this diagram to help you: A main program passing arguments into a function and receiving the return value

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **Function**
- ☐ **Procedure**
- ☐ **Parameter**
- ☐ **Variable**
- ☐ **Constant**
- ☐ **Sequence**

A A named subroutine that performs a task without returning a value.

B A value passed into a subroutine.

C A named subroutine that performs a task and returns a value.

D A named value that stays the same while the program runs.

E Statements executed one after another in order.

F A named memory location whose value can change while the program runs.

B Fill the gaps

Use the word bank · 5 marks

procedure | subroutine | reusable | function | parameters

A named block of code that performs one task is a _____. A _____ returns a value but a _____ does not. Values are passed in using _____. Writing code once and using it many times makes it _____.

C True or false?

Tick one box · 4 marks

	True	False
A procedure always returns a value.	☐	☐
Functions can be tested on their own.	☐	☐

Console.WriteLine() is a user-defined function.	☐	☐
Parameters allow data to be passed into a function.	☐	☐

D Exam-style questions

12 marks

1 Define the term function. [2]

2 Describe the difference between a function and a procedure. [2]

3 What is a parameter? [1]

4 Give two benefits of using user-defined functions. [4]

5 Write a C# function called Area that takes width and height as parameters and returns the area. [3]

E Quick quiz

Circle one answer · 5 marks

1 A subroutine that returns a value is a... [1]

A procedure B function
C loop D constant

2 In static int Total(int a, int b), a and b are... [1]

A return values B parameters
C constants D indexes

3 Which is a benefit of functions? [1]

A Code is repeated many times B Each part can be tested separately
C Programs run without a CPU D They remove the need for variables

4 Which keyword sends a value back from a C# function? [1]

A Console.WriteLine B return
C void D static

5 Math.Round() is an example of a... [1]

A user-defined function B built-in function

C procedure D class

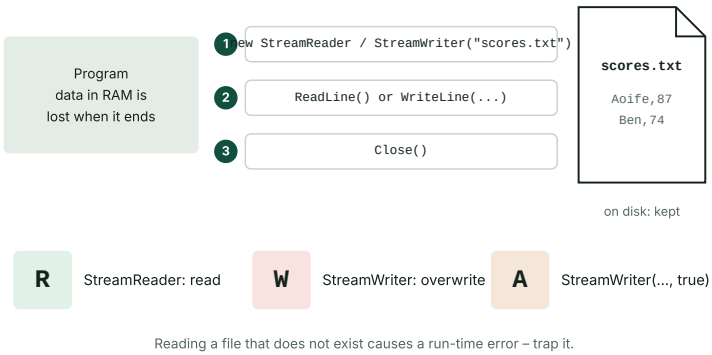
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · PROGRAMMING CONSTRUCTS

Basic file handling

D20 WORKSHEET

Name	Class	Date	Score

I can: Explain why programs store data in files · Open, write to, append to, read and close a text file · Process records read from a file



Use this diagram to help you: Opening, reading or writing, and closing a text file, with read, write and append modes

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **File handling**
- ☐ **Variable**
- ☐ **Constant**
- ☐ **Sequence**
- ☐ **Selection**
- ☐ **Iteration**

A Repeating a block of code (loops).

B A named memory location whose value can change while the program runs.

C A condition decides which statements run (IF statements).

D Opening, reading, writing, appending and closing files to store data permanently.

E A named value that stays the same while the program runs.

F Statements executed one after another in order.

B Fill the gaps

Use the word bank · 5 marks

closed | persistent | append | opened | Write

Files provide _____ storage so data is not lost when a program ends. A file must be _____ before use and _____ afterwards. _____ mode overwrites existing data but _____ mode adds to the end.

C True or false?

Tick one box · 4 marks

	True	False
Opening a file in write mode keeps the existing contents.	☐	☐
Files should be closed after use.	☐	☐

Reading a missing file causes a run-time error.	☐	☐
Data in RAM is kept after the program ends.	☐	☐

D Exam-style questions

13 marks

1 Explain why a program might store data in a file. [2]

2 Name and describe two file handling functions other than read and write. [4]

3 Describe the difference between write mode and append mode. [2]

4 Write the three steps needed to add a new score to an existing file. [3]

5 What type of error occurs if a program tries to read a file that does not exist? How can it be handled? [2]

E Quick quiz

Circle one answer · 5 marks

1 Which C# statement opens scores.txt to add data to the end? [1]

A new StreamReader("scores.txt")
B new StreamWriter("scores.txt")
C new StreamWriter("scores.txt", true)
D File.Delete("scores.txt")

2 Why are files used? [1]

A To make programs faster
B To store data permanently
C To translate code
D To create loops

3 What should happen after writing to a file? [1]

A Delete it
B Close it
C Compile it
D Reopen it in w mode

4 Opening an existing file with new StreamWriter("data.txt") will... [1]

A add to it
B overwrite it
C read it
D lock it forever

5 Which splits a CSV line into fields? [1]

A line.Split(',')

B line.Length

C sr.Close()

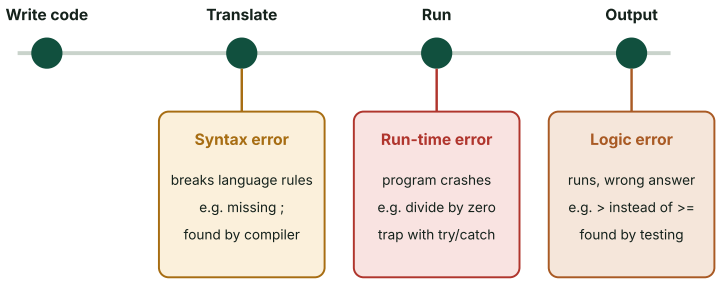
D line.ToUpper()

Validation, error types and error trapping

D21 WORKSHEET

Name	Class	Date	Score

I can: Describe presence, length, type, format and range checks and write validation loops · Distinguish syntax, execution (run-time) and logic errors and how each is found · Use try ... catch to trap errors



Dry runs, trace tables and breakpoints help find logic errors.

Use this diagram to help you: When syntax, run-time and logic errors appear during development

A Key terms

Write the letter of the matching definition · 4 marks

- ☐ **Syntax error**
- ☐ **Execution / run-time error**
- ☐ **Logic error**
- ☐ **Error trapping**

A Breaking the grammar rules of the language; found by the translator before running.

B An error that occurs while the program runs and causes it to crash, e.g. divide by zero.

C Code that detects a run-time error and handles it instead of crashing, e.g. try/except.

D The program runs but gives incorrect results because the reasoning is wrong.

B Fill the gaps

Use the word bank · 6 marks

syntax | Error trapping | logic | range | presence | execution

A _____ check makes sure a field is not empty. A _____ check makes sure a value is between limits. A _____ error breaks the rules of the language. A program that crashes while running has an _____ error. A program that runs but gives the wrong answer has a _____ error. _____ stops a program crashing.

C True or false?

Tick one box · 4 marks

	True	False
Validation can tell if data is correct.	☐	☐
Dividing by zero causes a run-time error.	☐	☐

A logic error stops the program from running.	☐	☐
A FOR loop is the best choice for a validation loop.	☐	☐

D Exam-style questions

22 marks

1 Explain why validation cannot guarantee that data is correct. [2]

2 Describe a format check and give an example in a password system. [2]

3 Write pseudocode that repeatedly asks for 'Y' or 'N' until one of them is entered, showing an error message for invalid input. [5]

4 Give one example each of a syntax error, an execution error and a logic error. [3]

5 Describe error trapping and one way it improves a program. [4]

6 Describe execution errors and logic errors and how each can be dealt with. (QWC) [6]

E Quick quiz

Circle one answer · 5 marks

1 A missing semicolon at the end of a C# statement is a... [1]

A logic error
B syntax error
C run-time error
D overflow

2 Which error is found by comparing output with expected results? [1]

A Syntax
B Logic
C Compile
D Type

3 Checking that a password is at least 8 characters is a... [1]

A presence check
B length check
C range check
D type check

4 Which C# construct traps run-time errors? [1]

A if/else
B try/catch
C foreach
D static/return

5 while (x < 1 || x > 10) is a validation loop for... [1]

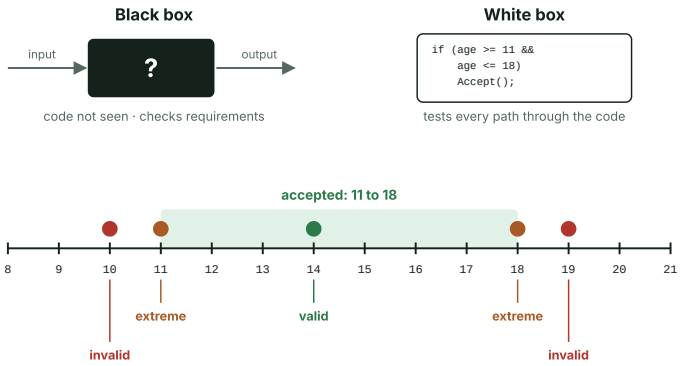
A 1 to 10 inclusive
B below 1 only
C over 10 only
D exactly 10

Testing approaches, test plans and test data

D22 WORKSHEET

Name	Class	Date	Score

I can: Describe white box, black box, unit, integration and system testing · Explain iterative testing · Create a test plan with valid, invalid, extreme and null test data



Also test invalid types ("ten") and null data (nothing entered).

Use this diagram to help you: Black box and white box testing, and test data on a number line for an 11 to 18 age range

A Key terms

Write the letter of the matching definition · 6 marks

- ☐ **White box testing**
- ☐ **Black box testing**
- ☐ **Unit testing**
- ☐ **Integration testing**
- ☐ **System testing**
- ☐ **Test plan**

A Testing the internal code, paths and logic, usually by the programmer.

B Testing a single module, function or method on its own.

C Testing the complete, integrated system against its specification.

D Testing inputs and outputs against requirements without seeing the code.

E Table listing each test, its purpose, test data, expected outcome, actual outcome and action.

F Testing that separately built units work together.

B Fill the gaps

Use the word bank · 5 marks

White box | Black box | extreme | unit | integration

_____ testing looks at the internal code. _____ testing checks inputs and outputs without seeing the code. Testing one module on its own is _____ testing. Checking modules work together is _____ testing. Data at the limits of a range is _____ data.

C True or false?

Tick one box · 4 marks

	True	False
Black box testing requires programming knowledge.	☐	☐

System testing checks the whole system against its requirements.	☐	☐
Null test data means entering nothing.	☐	☐
For a range 1–100, 100 is invalid data.	☐	☐

D Exam-style questions

17 marks

1 Describe two differences between black box and white box testing. [4]

2 Describe unit testing. [2]

3 Describe the difference between integration testing and system testing. [2]

4 A program accepts a number of frames from 10 to 40. Give one example each of valid, extreme and invalid test data. [3]

5 Complete a test plan row for testing a blank name entry: purpose, test data, type and expected outcome. [4]

6 Explain what is meant by iterative testing. [2]

E Quick quiz

Circle one answer · 5 marks

1 Testing every path through the code is... [1]

A black box B white box

C beta D alpha

2 For an age range 11–18, 18 is... [1]

A valid normal
B extreme
C invalid
D null

3 Testing modules work together is... [1]

A unit
B integration
C system
D acceptance

4 Which column records what actually happened? [1]

A Expected outcome
B Actual outcome
C Test data
D Purpose

5 Usual order of testing: [1]

A system → unit → integration
B unit → integration → system
C integration → system → unit
D system only

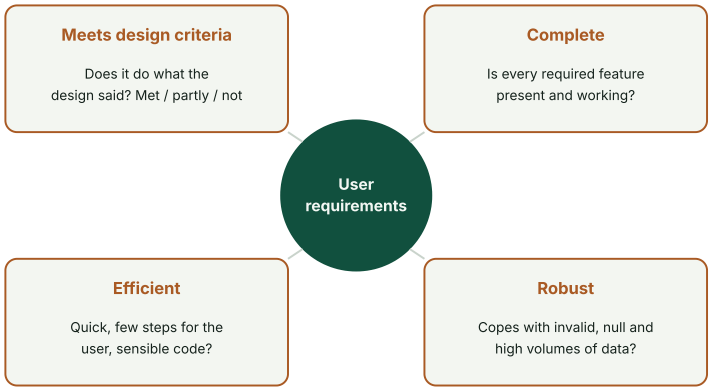
UNIT 4: DIGITAL DEVELOPMENT CONCEPTS · EVALUATION OF DIGITALLY AUTHORED SYSTEMS

Evaluating a solution

D23 WORKSHEET

Name	Class	Date	Score

I can: Explain the purpose of evaluation during and after development · Evaluate a solution against design criteria, completeness, efficiency and robustness · Use user requirements and test evidence in an evaluation



Use this diagram to help you: Evaluating a solution against the user requirements: meets design criteria, complete, efficient, robust

A Key terms

Write the letter of the matching definition · 4 marks

- ☐ **Evaluation**
- ☐ **User requirements**
- ☐ **Robustness**
- ☐ **Stakeholders**

A Judging the solution against the user requirements and design.

B The agreed list of what the system must do, used to judge the solution.

C People with an interest in the system: client, end users, developers.

D The ability to handle invalid, unexpected and high volumes of data without failing.

B Fill the gaps

Use the word bank · 5 marks

robust | user requirements | throughout | complete | efficient

Evaluation checks a solution against the original _____. A _____ solution has all required features working. A _____ solution copes with invalid data without crashing. An _____ solution does not waste time or resources. Evaluation should take place _____ development.

C True or false?

Tick one box · 4 marks

	True	False
Evaluation only happens after the program is finished.	☐	☐
Robustness testing should use high volumes of data.	☐	☐
A list of features is the same as an evaluation.	☐	☐

Stakeholders include the end users of the system. ☐ ☐

D Exam-style questions

10 marks

1 Give two reasons why a solution should be evaluated. [2]

2 Explain how user requirements are used during evaluation. [2]

3 What is meant by a robust solution? [2]

4 Describe how a developer could test the robustness of a booking system. [2]

5 Explain what is meant by an efficient solution. [2]

E Quick quiz

Circle one answer · 5 marks

1 Evaluation measures a solution against... [1]

A the CPU speed
B the user requirements
C the number of lines of code
D the font used

2 A solution that copes with invalid input without crashing is... [1]

A efficient
B robust
C complete
D abstract

3 Testing with very large amounts of data checks... [1]

A syntax
B robustness
C abstraction
D compilation

4 A complete solution... [1]

A has every required feature working
B has no comments
C uses only FOR loops
D is written in Java

5 Which is an evaluative comment? [1]

A The program has a menu.
B Requirement 2 is fully met: all 10 test bookings were stored correctly.
C I used C#.
D The program is in a file.